

Aditya Khandelwal

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PROFESSIONAL SUMMARY

Execution-first CS undergraduate who builds working AI-powered systems end-to-end across any tech stack. Across 3 internships, built LLM extraction pipelines at Sentient Scripts, researched time series forecasting at Samsung R&D, and developed real-time CV systems at Infosys. Independently built multi-agent (PromptGraph) and full-stack RAG (VectorLearn) systems. Uses AI as leverage to move fast and deliver outcomes.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI / ML: LangGraph, LangChain, OpenAI API, Groq API, Gemini API, RAG, FAISS, PyTorch, YOLOv8, OpenCV

Backend: FastAPI, REST APIs, SSE Streaming, Pydantic, asyncio, Microservices

Full-Stack: Next.js, React.js, Vite, Express.js

Databases & DevOps: PostgreSQL, MySQL, SQLite, FAISS, Git, Docker, CI/CD, Linux

PROFESSIONAL EXPERIENCE

- **AI Engineer Intern** Aug 2025 – Mar 2026
Sentient Scripts, Discovery Lab SRMIST
 - Built multi-source data aggregation pipelines in Python that ingested, normalized, and consolidated intelligence from 5+ external feeds for real-time monitoring and trend tracking.
 - Developed an LLM-based extraction module that transformed unstructured text into structured domain-specific metadata, reducing manual extraction time by over 70% and enabling automated cross-dataset correlation.
 - Designed an end-to-end intelligence pipeline integrating scheduled data collection, LLM-driven correlation with Pydantic-validated output, and automated report generation, cutting turnaround from hours to minutes.
- **Research Intern** Apr 2025 – Jan 2026
Samsung R&D Institute India Bangalore (Remote)
 - Benchmarked the TOTEM time series forecasting model with PyTorch across 4+ datasets and multiple forecasting horizons, producing comparative accuracy analyses against baseline transformer architectures.
 - Identified tokenization and architecture design trade-offs that impacted long-horizon generalization, contributing findings to the team's ongoing model evaluation process.
- **Python Developer Intern** Nov 2025 – Jan 2026
Infosys Springboard (Remote)
 - Built a multi-zone real-time crowd counting system using YOLOv8 and OpenCV with Python multiprocessing to concurrently handle 4+ video streams, achieving measurable per-frame latency reduction.
 - Designed a centralized monitoring dashboard with automated threshold-based alerting, eliminating manual stream supervision and reducing incident response time across all monitored zones.

TECHNICAL PROJECTS

- **PromptGraph - Multi-Agent Prompt Architecture Generator** [\[GitHub\]](#)
Tech Stack: Python, LangGraph, Groq API, Pydantic, FastAPI
 - Architected a 6-agent LangGraph StateGraph pipeline that converts natural language project ideas into structured prompt architectures and production-ready Python scaffolding in a single run.
 - Implemented a 3-layer fault-tolerance system (Tenacity exponential backoff, rate limiting, schema self-healing) that reduced token consumption by ~59% via compact schema representation.
 - Built a Critic-Refiner quality loop evaluating prompts across 5 dimensions; validated with a 190-test suite covering unit, integration, and live API regression.

- **VectorLearn - Document Intelligence Platform** [\[GitHub\]](#)
Tech Stack: Python, FastAPI, React, Vite, TailwindCSS, FAISS, SQLite, Ollama, RAG
 - Built a full-stack AI tutoring platform with a RAG pipeline supporting multi-user document upload, isolated FAISS vector stores per user, and an SSE-streaming chat interface returning source-cited answers.
 - Engineered a modular `docpipe` ingestion pipeline handling PDF extraction, recursive chunking, `bge-large` embeddings via Ollama, and per-user FAISS+SQLite indexing for accurate semantic retrieval.
- **AI PPT Generator - Real-Time Presentation Deck Builder** [\[GitHub\]](#)
Tech Stack: Python, FastAPI, Groq API, Vite, JavaScript, SSE, pptxgenjs
 - Built a prompt-to-deck system powered by Groq API that streams slide content progressively via SSE with in-browser editing and reordering, cutting deck creation time to under 60 seconds.
 - Designed a FastAPI backend with typed SSE event contracts (title/slide/done/error) and a Node.js `pptxgenjs` export script; supported configurable tone, slide count (5-20), and multi-language output.

HACKATHONS & ACHIEVEMENTS

- **Code Zero Hackathon - 2nd Place** *National Level*
 - Outperformed teams nationwide by designing and building a functional AI prototype under strict time constraints, demonstrating execution speed and problem-solving under pressure.

EDUCATION

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| • SRM Institute of Science and Technology | 2023 – 2027 |
| B.Tech in Computer Science and Engineering | CGPA: 8.91 / 10 |
| • BVB Vidyashram | |
| Senior Secondary (Class XII) | 82.80% |